Introduction to the nucleus:

* First discovered by Rutherford in 1911.
* Dimension of nucleus is in the order of 10^{-14} m.
* Nucleus consists of protons and neutrons and surrounded by electrons in the orbit
* A species of nucleus is known as nuclide and represented as $_{Z}X^{A}$

where Z = Atomic number
A = mass number
X = chemical symbol of the species.

e.g. $_{17}Cl^{35}$, Z = 17 protons
A = 35 nucleons (proton + neutrons).
N = neutrons = A − Z = 35 − 17 = 18.

Classification of Nuclei:

Atoms of different elements are classified as:

(a) Isotopes: Same Atomic number Z and different atomic mass.
e.g. $_{14}Si^{28}$, $_{14}Si^{30}$, $_{14}Si^{32}$ etc.

(b). Isobars: Same mass number A, but different atomic number Z.
e.g. $_{8}O^{16}$ & $_{7}N^{16}$,

(c). Isotones: Nuclei with equal number of neutrons.
e.g. $_{6}C^{14}$, $_{7}N^{15}$, $_{8}O^{16}$, [N = A − Z = 8]

(d) Isomers: Same number of A & Z with different energy states.

(e) Mirror nuclei: Same number of A but proton and neutron numbers are interchangable.
e.g. $_{4}Be^{7}$ $(Z=4, N=3)$, $_{3}Li^{7}$ $(Z=3, N=4)$.

General properties of Nucleus:

(1). Nuclear size:

From Rutherford's α-particle scattering experiment the radius of atomic nucleus is of the order of 10^{-14} m to 10^{-15} m, while that of an atom is 10^{-10} m. The empirical formula for the nuclear radius is

$$R = r_0 A^{1/3}.$$

where A = mass number

$$r_0 = 1.3 \times 10^{-15}\ m = 1.3\ fm\ (fermi).$$

$(\because 1\ fm = 1\ fermi = 10^{-15}\ m)$.

From this formula radius of any atom can be calculated.

e.g. Radius of $_{6}C^{12}$ nucleus, $R = (1.3)(12)^{1/3} = 3$ fm

Radius of $_{47}Ag^{107}$ nucleus, $R = (1.3)(107)^{1/3} = 6.2$ fm. etc.

The nuclear radius may be estimated from the scattering of neutrons and electrons by the nucleus. There may be other methods as well.

The scattering of fast electrons with high energy (10^4 MeV) are alse employed to measure charge density.

(2) Nuclear Mass:

The nuclear mass is the sum of mass of protons and mass of neutrons. Then mass of nucleus

$$M = Zm_p + Nm_n$$

where, m_p = mass of a proton

m_n = mass of a neutron.

The nuclear mass can be measured by mass spectrometers. It has revealed experimentally that the real nuclear mass is always less than the sum of mass of individual nucleons.

i.e. real nuclear mass $< Zm_p + Nm_n$.

The difference in mass $\Delta m = (Zm_p + Nm_n) - \text{real mass}$.

This difference is called mass defect. This mass is utilized to the stability of nucleus, binding the nucleons together.

(3) Nuclear Density:

As in general concept, nuclear density is the ratio of nuclear mass to its volume. It is denoted as ρ_N. $\therefore \rho_N = \frac{\text{Nuclear mass}}{\text{Nuclear Volume}}$.

The total nuclear mass $= Am_N$

where A = mass number

m_N = mass of a nucleon $= 1.67 \times 10^{-27}$ kg.

Then nuclear volume $= \frac{4}{3}\pi R^3$

$$= \frac{4}{3}\pi (r_0 A^{1/3})^3 = \frac{4}{3}\pi r_0^3 A.$$

$$\rho_N = \frac{Am_N}{\frac{4}{3}\pi r_0^3 A} = \frac{m_N}{\frac{4}{3}\pi r_0^3} = \frac{1.67 \times 10^{-27}}{\frac{4}{3}\pi (1.3 \times 10^{-15})^3} = 1.816 \times 10^{17} \text{ Kg m}^{-3}.$$

It is tremendously high value of density of nucleus. Thus nuclear matter is highly compressed. Celestial objects like white dwarfs are having density nearly to nuclear density.

(4) Nuclear charge:

An atomic nucleus is composed of protons & neutrons. The neutrons are neutral particles, so charge of nucleus is solely due to protons. Each proton has a positive charge of 1.6×10^{-19} C. The nuclear charge is Ze, where Z is atomic number of the nucleus.

(5) Spin angular momentum:

Likewise in electron, both proton and neutron have an intrinsic spin. The spin angular momentum is computed as

$$L_s = \sqrt{s(s+1)}\,\frac{h}{2\pi}, \quad [s = \text{spin quantum no}]$$

The spin value is commonly taken as $\frac{1}{2}$.

Then $$L_s = \sqrt{\tfrac{1}{2}\cdot(\tfrac{1}{2}+1)}\,\frac{h}{2\pi} = \frac{\sqrt{3}}{2}\,\frac{h}{2\pi}.$$

(6) Nuclear magnetic dipole moments:

All the elementary particles proton, neutron & electrons are possessing spin. They all have magnetic dipole moment. proton has positive elementary charge and due to its spin, it should have a magnetic dipole moment $\mu_N = \frac{eh/2\pi}{2m_p}$

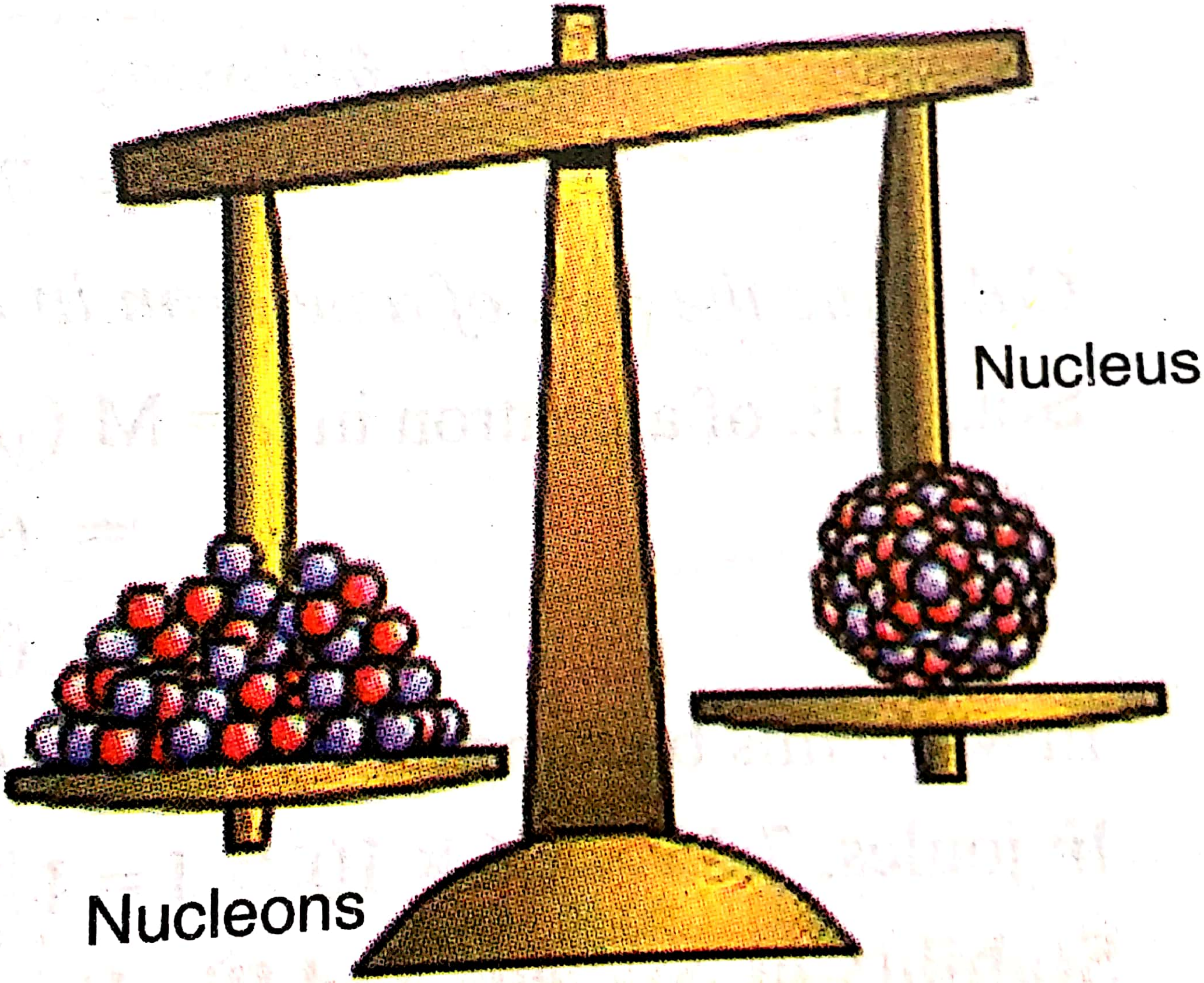
where m_p = mass of proton
μ_N = nuclear magneton.
Nuclear magnetic moment $\mu_N = 5.050 \times 10^{-27}$ J/T.

(7). Electric quadrupole moment:
In addition to its magnetic moment, a nucleus may have an electric quadrupole moment. An electric quadrupole moment is zero for atoms & nuclei in stationary states.
The nuclear electric quadrupole moment is a parameter which describes the effective shape of the ellipsoid of nuclear charge distribution.

Binding Energy:

The theoretical explanation for the mass defect is based on Einstein's equation $E = mc^2$. When Z protons & N neutrons combine to make a nucleus, some of the mass (Δm) disappears. This mass converts into an amount of energy $\Delta E = \Delta m \cdot c^2$. This energy is called Binding Energy (B.E.) of the nucleus. More the binding energy, more stable will be the nucleus.

To disrupt/break a stable nucleus into its constituent parts i.e. protons & neutrons, the minimum energy required is the binding energy. The magnitude of B.E. of a nucleus determines its stability against disintegration. If the B.E. is large, the nucleus is stable. A



Binding Energy.

nucleus having the least possible energy equal to the B.E, is said to be in the ground state. If $E > E_{min}$, it is said to be in the excited state. The case $E = 0$ corresponds to dissociation of the nucleus into its constituent nucleons.

The binding energy is expressed as

$$B.E = \{(Zm_p + Nm_n) - M\}c^2 \simeq \Delta m c^2.$$

where M = experimentally determined mass of a nucleide having Z protons and N neutrons.

Following may be the cases.

I/ If $B.E > 0$, the nucleus is stable

II/ If $B.E < 0$, the nucleus is unstable.

Illustration:

Nucleon: Deuteron.

Constituents: 1 proton + 1 neutron.

$m_p = 1.007276$ amu

$m_n \doteq 1.008665$ amu.

Mass of proton + neutron (free state) = 2.015941 amu.

Mass of Deuteron nucleus = 2.013553 amu

Mass defect $\Delta m = 0.002388$ amu.

Binding energy $\Delta E = \Delta m \times 931$ MeV

$= 0.002388 \times 931$ MeV

$= 2.23$ MeV. ($\because$ 1 amu = 931 MeV)

Explanation:

When a deuteron is formed from a free proton & neutron, 2·23 MeV of energy is liberated. On the other hand 2·23 MeV energy must be supplied from an external source to break a deuteron up into a proton & a neutron. A γ-ray photon of 2·23 MeV energy can split deuteron into constituent parts.

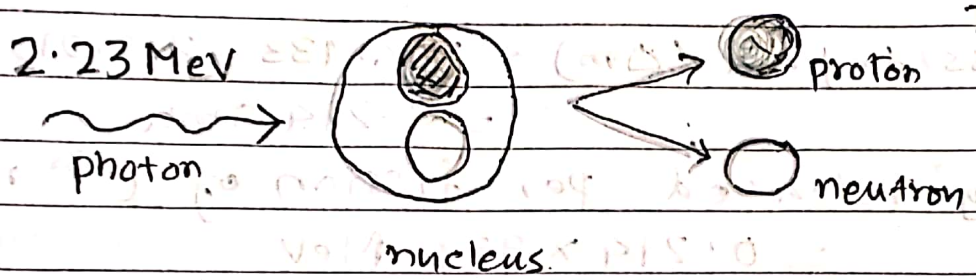


fig. Breaking of Deuteron nucleus

Alternatly

Mass defect $(\Delta m) = 0.002388$ amu

$= 0.002388 \times 1.66 \times 10^{-27}$ kg.

$c = 3 \times 10^{8}$ m/s.

$$B.E = \Delta m c^2$$

$$= 0.002388 \times 1.66 \times 10^{-27} \times (3 \times 10^{8})^2$$

$$= 3.5676 \times 10^{-13} \text{ J}$$

$$= \frac{3.5676 \times 10^{-13}}{1.6 \times 10^{-19}} = 2.2297 \times 10^{6} \text{ eV}$$

$$= \frac{2.2297 \times 10^{6}}{10^{6}} = 2.23 \text{ MeV}$$

$$= 2.23 \text{ MeV}.$$

Calculate the energy released in the following nuclear reaction:

$${}_{92}U^{235} + {}_{0}n^{1} \rightarrow \left[{}_{92}U^{236}\right] \rightarrow {}_{42}Mo^{95} + {}_{57}La^{139} + 2\,{}_{0}n^{1} + \beta^{*}$$

(* Emission of β-particle is neglected)

Mass of U^{235} nuclide $= 235.124$ amu
mass of one neutron $= 1.009$ amu.
Total mass $= 236.133$ amu

Mass of Mo^{95} $= 94.946$ amu
Mass of La^{139} $= 138.955$ amu
Mass of 2 neutrons $= 2.018$ amu.
Total mass $= 235.919$ amu

Now mass defect $(\Delta m) = 236.133 - 235.919$
$= 0.214$ amu

The energy released per fission of U^{235} nucleus
$= 0.214 \times 931$ MeV
$= 200$ MeV.

[Mo = Molybdenum, La = Lanthanum]

Stability of nucleus & binding energy:

The binding energy per nucleon can be obtained as

$$\text{B.E per nucleon} = \frac{\text{Total binding energy of the nucleus}}{\text{The number of nucleons it contains.}}$$

The binding energy per nucleon is plotted as a function of mass number A. The curve rises steeply at first and then more gradually until it reaches a maximum of 8.79 MeV at A=56, corresponding to Iron nucleus $_{26}Fe^{56}$. The curve then drops slowly to about 7.6 MeV at the highest mass number. The nuclei of intermediate

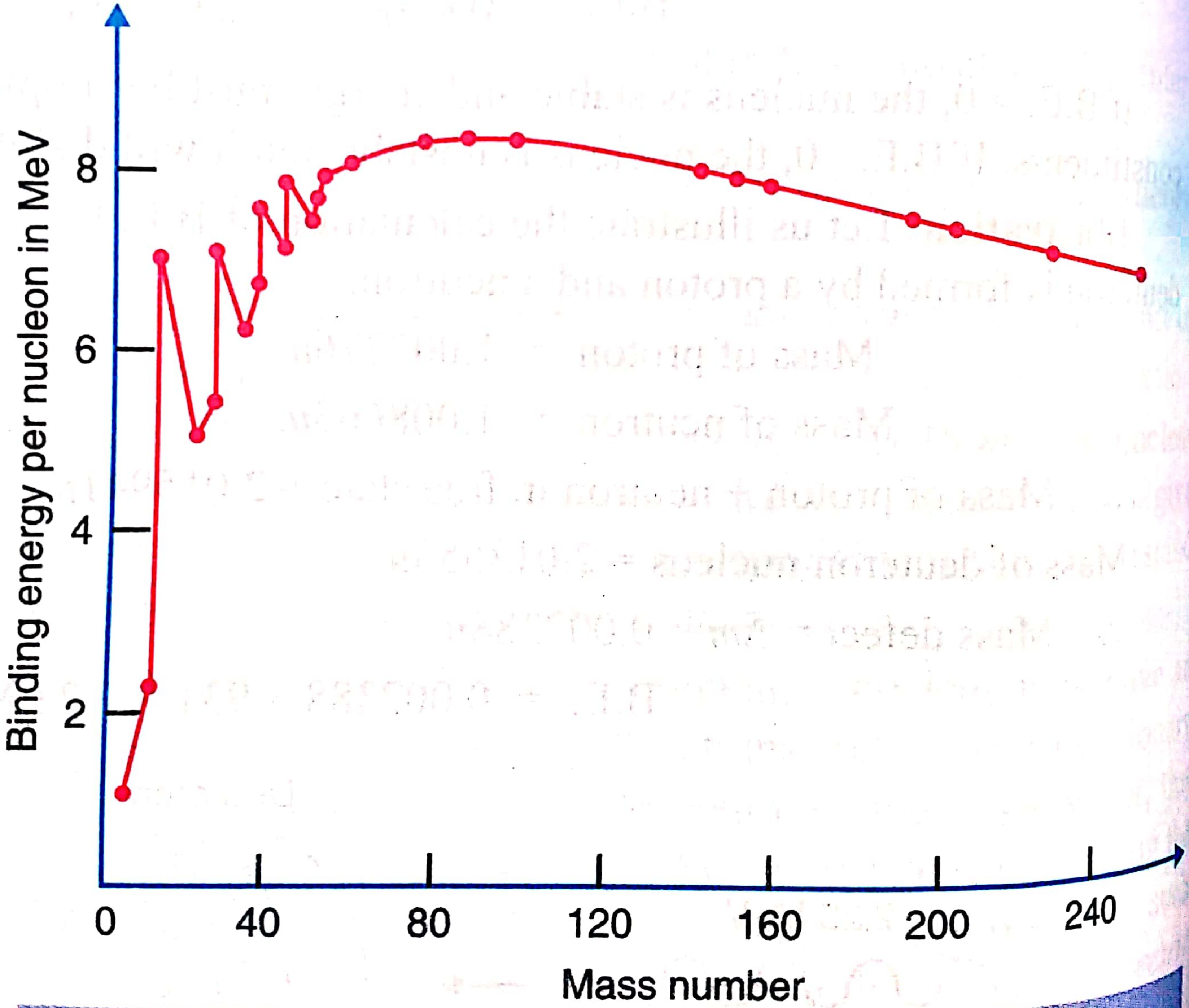

Binding energy per nucleon in MeV
8
6
4
2
0
40
80
120
160
200
240
Mass number

Fig. 27.3

mass are the most stable, since the greatest amount of energy is required to supply to liberate each of their nucleons.

fig 27.3

Packing fraction:

The ratio between the mass defect (Δm) and the mass number (A) is called the Packing fraction (f).

$$\therefore f = \Delta m / A$$

The mass defect per unit nucleon is called packing fraction. Atomic masses are measured relative to C-12, the packing fraction for this isotop is zero. Packing fraction also is a measure of the comparative stability of the atom. It is also defined as,

$$\text{Packing fraction} = \left(\frac{\text{Isotopic mass} - \text{Mass number}}{\text{Mass number}} \right) \times 10^4.$$

The packing fraction may have a negative or positive value.

i) If packing fraction is negative the isotopic mass is less than the mass number. The mass gets transferred into energy in the formation of the nucleus. ($E = mc^2$). These types of nuclei are more stable.

ii) If packing fraction is positive mass number is lesser than the isotopic mass. It would imply a tendency towards instability of the atom.

fig. 27.4

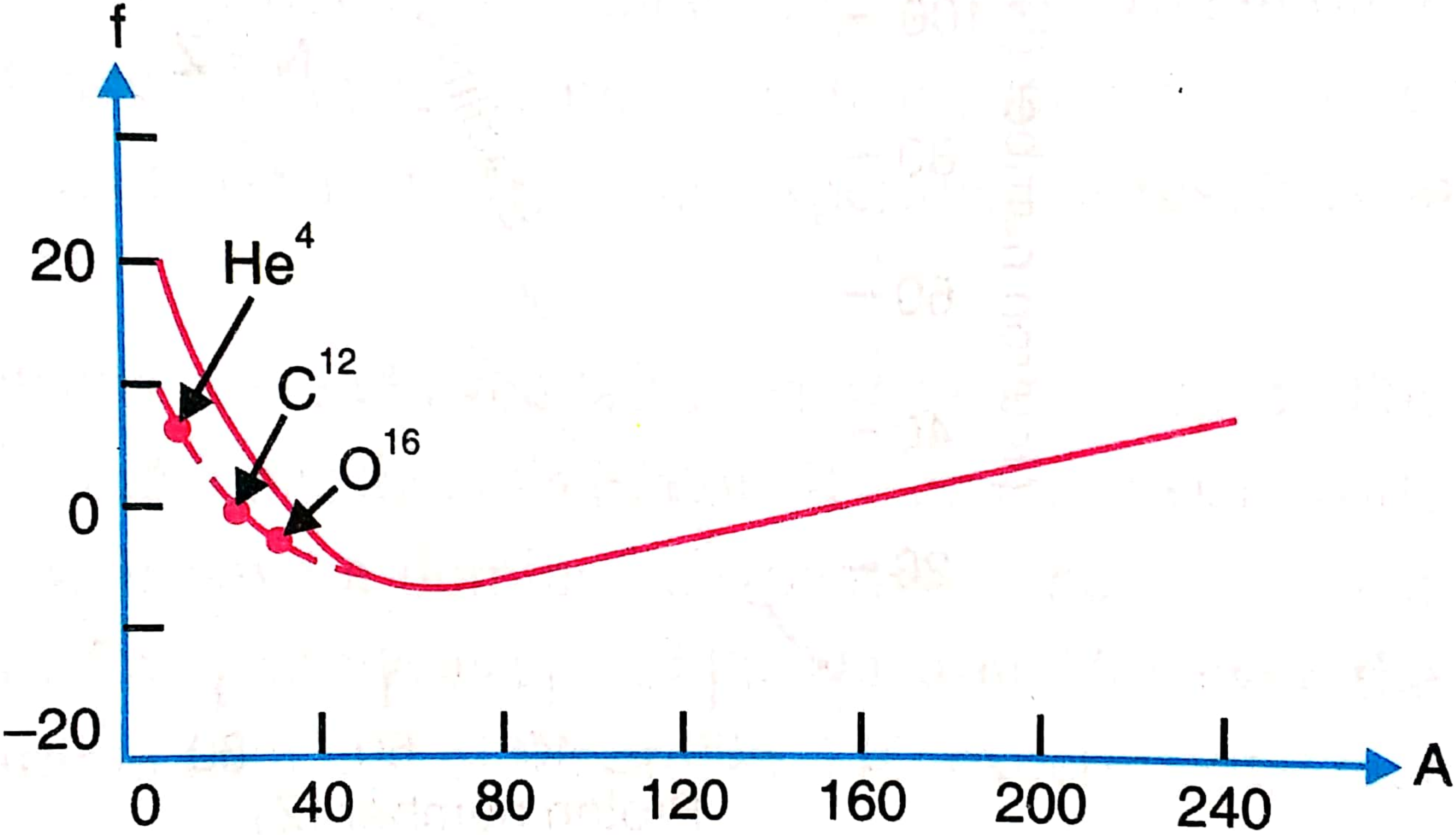


Fig. 27.4

In the plot of packing fraction against the atomic mass number, it gradually falls and reaches to a maximum negative value around A=45. Then after there is gradual increase and becomes positive beyond A=200. However He, C and O of mass number 4, 12 and 16 respectively do not fall on this curve. They are having smaller value and are more stable. The ements with mass numbers beyond 230 are radioactive and undergo disintegration spontaneously.

Nuclear Stability:

In nature there are different kinds on nuclei on the basis of their nucleons protons and neutrons having even-even, odd-odd, even-odd or odd-even ~~ee~~ numbers of composition. The stable 272 stable nuclei can be classified as in the following table.

The combination of even number of proton and even number of neutron is preferred to be most stable by nature. It is 160 in number, The even-odd and odd-even combination of proton-neutron nuclei are both nearly equal. whereas combination of odd-odd proton-neutron is least

S.N.	Protons	Neutrons	Stable nuclides
1	even	even	160
2	even	odd	56
3	odd	even	52
4	odd	odd	4
		Total	272

$\boxed{\text{Fig 27.5 \& 27.6}}$

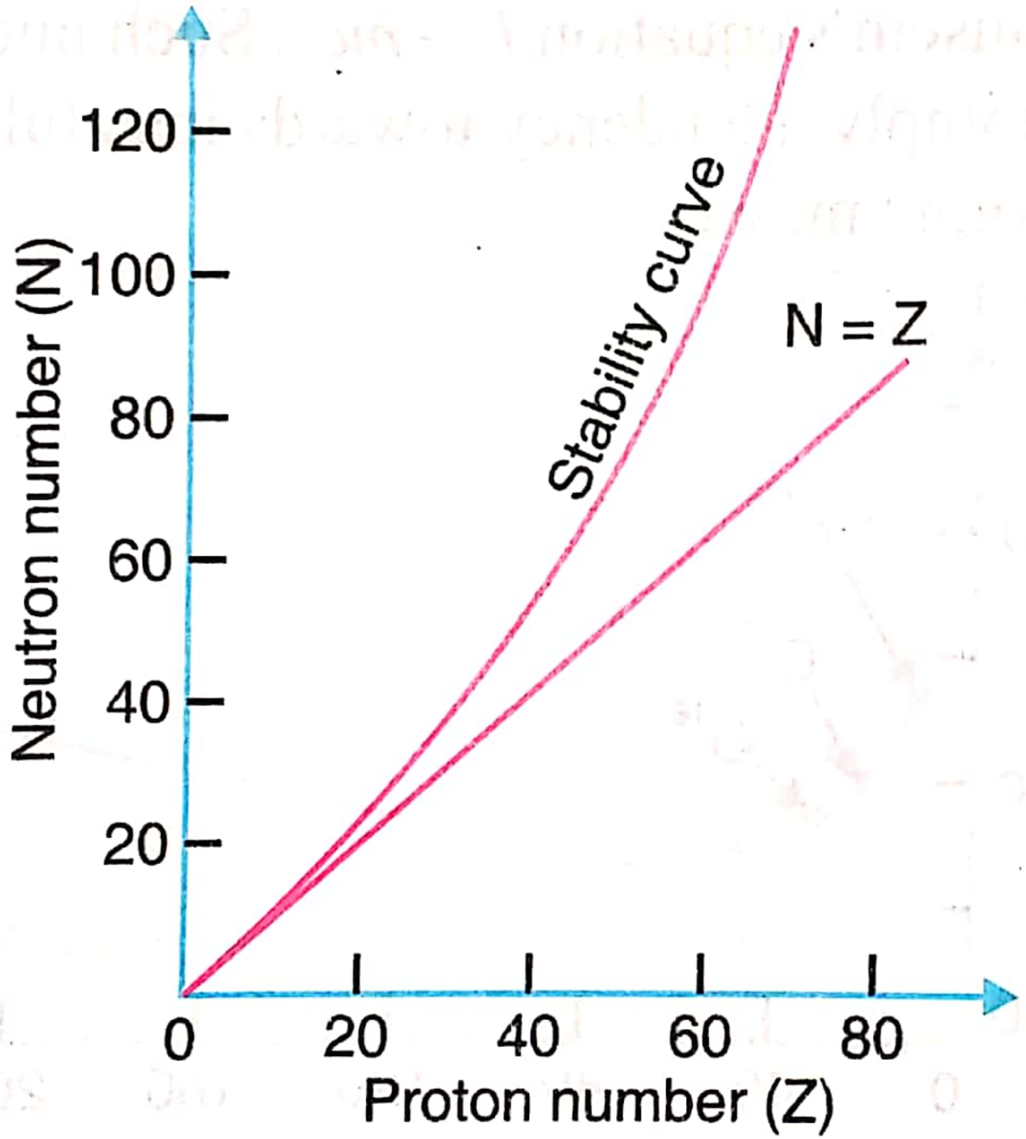


Fig. 27.5

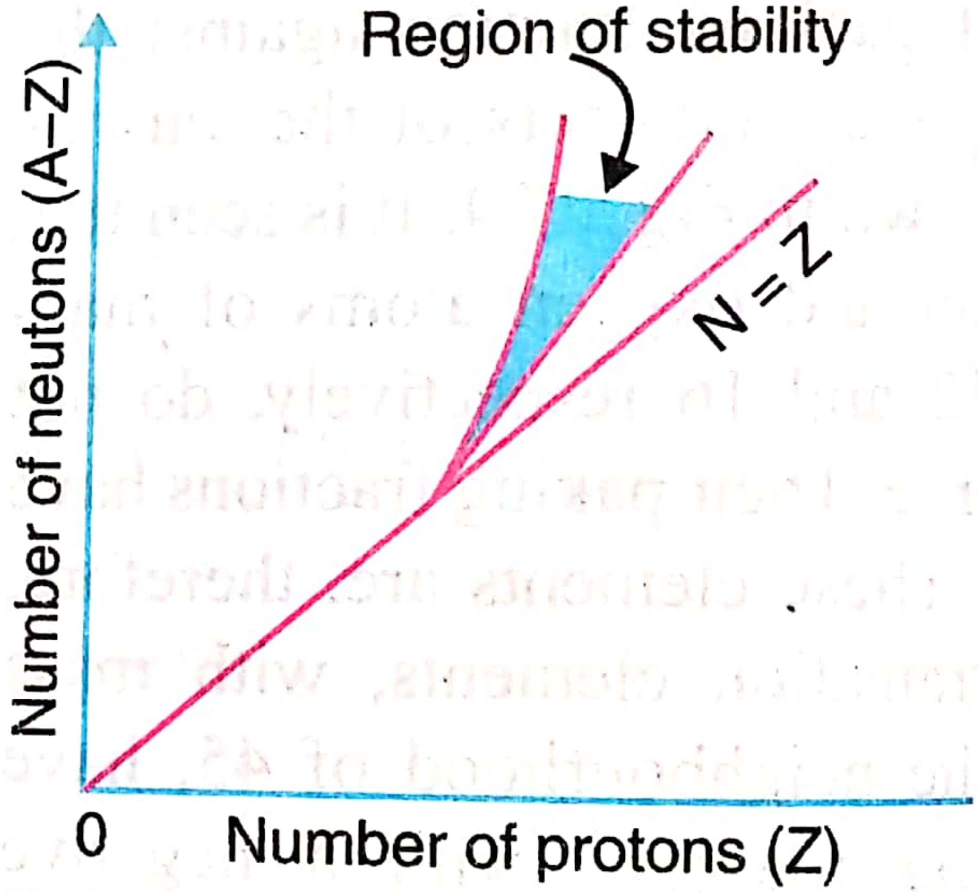


Fig. 27.6

preferred by nature, which simply 4 in number.

In the plot of number of neutrons Vs number of protons for the stable nuclides shows that, for $Z<20$, the stability line is straight with $Z=N$. For $Z>20$ & $N>20$ the stability curve bends in the direction of $N>Z$. For example, $_{20}Ca^{48}$, has $N=28$ and $Z=20$. Likewise for $_{91}Pa^{232}$ (protactinium) which as $N=141$ & $Z=91$. The larger value of N signifies towards the stability of the nucleus. For large value of Z Coulomb electrostatic repulsion among the protons becomes important. To reduce the force the number neutrons must be greater.

On plotting neutron number versus proton number (N−Z), all the stable nuclei fall within the shaded region of stability curve. The nuclei above & below the shadded region are unstable.

Theory of Nuclear Composition:-

Among the several proposed theories of nuclear composition, there are two prominent theories-

i) Proton-electron hypothesis
ii) Proton-neutron hypothesis.

Proton-electron hypothesis:

Peoples were observing emission of −vely charged particle (β) from the nucleus. They thought that nucleus should contain proton and electron in the nucleus. According to this hypothesis a nucleus of mass number A ~~(239)~~ and atomic number Z contains A protons and A−Z electrons. The nucleus is surrounded by Z electrons, so that the atom is electrically neutral.

Example: In a Sodium $_{11}Na^{23}$ atom/nucleus, in it's nucleus there are 23 protons (A) and (23−11) = 12 (A−Z) electrons. So nucleus of Sodium consists of 23 protons & 12 electrons. The 11 unit of positive charge in the nucleus is balanced by 11 electrons in the orbit.

Why electrons cannot exist inside the nucleus?

The dimension of typical nuclei are in the order of 10^{-14} m (radius). If an electron exists inside a nucleus, the uncertainity in its position (Δx) may not exceed 10^{-14} m. According to uncertainity principle, the uncertainity in the electrons momentum is

$$\Delta p \geq \frac{\hbar}{\Delta x} \quad (\text{where } \hbar = \frac{h}{2\pi})$$

$$\text{or } \Delta p \geq \frac{1{\cdot}054 \times 10^{-34}}{10^{-14}} \geq 1{\cdot}1 \times 10^{-20} \text{ kg m s}^{-1}.$$

If this is the uncertainity in the momentum of electron, the momentum itself must be at least comparable in magnitude.

The approximate momentum of the electron $= 1{\cdot}1 \times 10^{-20}$ kgms^{-1}

An electron whose momentum is this much value has a kinetic energy many times greater than its rest energy m_0c^2, i.e. $T(K.E.) >> m_0c^2$

From relativistic formula $T = pc$.

$$\text{or} \quad T = (1.1\times10^{-20})(3\times10^{8})$$
$$= 3.3\times10^{-12}\ \text{J}.$$
$$= \frac{3.3\times10^{-12}}{1.6\times10^{-13}}\ \text{MeV} = 20.63\ \text{MeV}.$$

Thus nuclear electron must have K.E more than 20 MeV. But maximum energy possessed by electron is found only 2 to 3 MeV. Hence it can be concluded that electron can not exist in the nucleus.

Proton - Neutron hypothesis:

After the discovery of neutron by Chadwick in 1932, proton-neutron theory has been accepted. According to this, a nucleus of atomic number Z and mass number A consists of Z protons & A-Z neutrons. The number of electrons in the extranuclear space is also Z, so that atom as a whole is neutral.

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